LESSON PLAN

AVIONICS AND ELECTRONIC WARFARE (13)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	04
Total No. of Lesson Plans	25	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Batteries- functions, different types of batteries used in A/C, Location of batteries in A/C battery connection, typical battery systems, on
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Batteries- functions, different types of batteries used in A/C, Location of batteries in A/C battery connection, typical battery systems, on board battery charging and warning system and memory effect on NI- Cd batteries. Practical in battery shop. Safety precautions to be observed in battery shop.
5. Specific Objectives	13.1.1.1 Trainee will be able to explain the A/C power systems.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Practical in battery shop. Safety precautions to be observed in battery shop. Instructor supervision during handling and inspection.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a battery in an aircraft?
(b) Name two common types of batteries used in aviation.
(c) Why is it critical to locate batteries correctly in an aircraft?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Functions of aircraft batteries • Types of batteries used in A/C • Location of batteries in A/C • Battery connection points • Typical battery systems	L & Ds	PPT, IB	15 min
(b)	• On board battery charging systems • Warning system indicators • Memory effect in Ni-Cd batteries • Maintenance of Ni-Cd batteries • Troubleshooting battery faults	L & Ds	PPT, PDF	20 min
(c)	• Practical in battery shop • Safety precautions in battery shop • Instructor supervision protocols • Inspection of battery terminals • Verification of charging systems	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the memory effect observed in Ni-Cd batteries.
(b) Describe the safety precautions required when handling batteries in the shop.
(c) Identify the typical location of batteries in an aircraft and their connection points.

12. Home Assignment and Follow Up	Prepare a brief report on the differences between lead-acid and Ni-Cd batteries used in aircraft, referencing the next lesson on constant frequency systems.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	A/C power supply systems-Constant frequency systems, operation of CSDU, integrated drive generators, field excitation of frequency wild generators.	Refer to the master syllabus reference list	LP 05

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min